

ASP.NET Core MVC CRUD with SQL Server (Complete Steps)

Step 1: Create a New Project

1. Open **Visual Studio 2022/2026**
2. Click **Create a new project**
3. Select **ASP.NET Core Web App (Model-View-Controller)**
4. Click **Next**

Project Name:

Emp_Management

Location:

D:\net\

Click **Next**

Choose

.NET 9

Authentication

None

Check

Configure for HTTPS ✓

Click **Create**

Step 2: Install NuGet Packages

Right-click the project

Manage NuGet Packages

Install

Microsoft.EntityFrameworkCore.SqlServer

Install

Microsoft.EntityFrameworkCore.Tools

Your .csproj should contain

```
<ItemGroup>
```

```
  <PackageReference Include="Microsoft.EntityFrameworkCore.SqlServer" Version="9.0.0" />
```

```
  <PackageReference Include="Microsoft.EntityFrameworkCore.Tools" Version="9.0.0">
```

```
    <PrivateAssets>all</PrivateAssets>
```

Step 3: Create SQL Server Database

Open SQL Server Management Studio.

Run

```
CREATE DATABASE EmployeeDB1;
```

```
GO
```

Refresh Databases.

Step 4: Create Folder Structure

Create these folders.

Controllers

Models

Data

Views

Inside Views create

Employee

Step 5: Create Employee Model

Models

Employee.cs

```
using System.ComponentModel.DataAnnotations;
```

```
namespace Emp_Management.Models
```

```
{
```

```
public class Employee
{
    public int Id { get; set; }

    public string? Name { get; set; }

    public string? Department { get; set; }

    public string? Email { get; set; }
}
}
```

Step 6: Create DbContext

Inside

Data

Create

AppDbContext.cs

```
using Microsoft.EntityFrameworkCore;
```

```
using Emp_Management.Models;
```

```
namespace Emp_Management.Data
```

```
{
```

```
    public class AppDbContext : DbContext
```

```
    {
```

```
        public AppDbContext(DbContextOptions<AppDbContext> options)
```

```
            : base(options)
```

```
{  
}  
  
    public DbSet<Employee> Employees { get; set; }  
}  
}
```

Step 7: Connection String

Open

appsettings.json

```
{  
  "ConnectionStrings": {  
    "DefaultConnection":  
    "Server=.\SQLEXPRESS;Database=EmployeeDB1;Trusted_Connection=True;TrustServerCertificate=True;"  
  }  
}
```

If your instance is

SQLEXPRESS01

then

```
Server=.\SQLEXPRESS01;
```

Step 8: Configure Program.cs

Replace with

```
using Emp_Management.Data;
```

```
using Microsoft.EntityFrameworkCore;
```

```
var builder = WebApplication.CreateBuilder(args);

builder.Services.AddControllersWithViews();

builder.Services.AddDbContext<AppDbContext>(options =>
    options.UseSqlServer(
        builder.Configuration.GetConnectionString("DefaultConnection")));

var app = builder.Build();

if (!app.Environment.IsDevelopment())
{
    app.UseExceptionHandler("/Home/Error");
    app.UseHsts();
}

app.UseHttpsRedirection();

app.UseStaticFiles();

app.UseRouting();

app.UseAuthorization();

app.MapControllerRoute(
    name: "default",
```

```
pattern: "{controller=Employee}/{action=Index}/{id?}");
```

```
app.Run();
```

Step 9: Create Migration

Open

Tools



NuGet Package Manager



Package Manager Console

Run

Add-Migration InitialCreate

Step 10: Update Database

Run

Update-Database

SQL Server creates

EmployeeDB1



Employees

table.

Step 11: Scaffold Controller

Right-click

Controllers



Add



New Scaffolded Item



MVC Controller with views, using Entity Framework

Choose

Model Class

Employee

Choose

Data Context

AppDbContext

Controller Name

EmployeeController

Click

Add

Visual Studio automatically creates:

EmployeeController.cs

and inside

Views/Employee

Create.cshtml

Delete.cshtml

Details.cshtml

Edit.cshtml

Index.cshtml

Step 12: Run the Project

Press

F5

or

Ctrl + F5